Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)			award (1) for conditions under which both conduct electricity barium oxide only conducts electricity when molten (or in solution) whilst barium conducts electricity when (molten or) solid the ions in solid barium oxide cannot move freely but in the molten state they are free to move (and carry electricity) (1) barium has delocalised electrons in the solid state (which can carry electricity) (1)	3			3		
	(d)			0.135 nm (1) award (1) for either of following • radius considerably reduced since 2 electrons lost (to form ion) • Ba^{2+} has one less shell of electrons award (1) for correct explanation if 0.210 nm chosen		1	1	2		
	(e)			$137.3 = \frac{134.9x + 137.9(100-x)}{100}$ (1) $-60 = -3x$ $x = 20$ (1) accept alternative methods and reasoning e.g. difference in mass of isotopes is 3.0 and A_r value is 2.4 more than lighter isotope and 0.6 less than heavier isotope ⇨ must be 80% of the heavier isotope		2		2	2	
				Question 10 total	**3**	**7**	**5**	**15**	**4**	**3**